Binary Classification of AOI Alarms into True Defects and False Calls

Second check of AOI results with supervised machine learning (Siemens production data)

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Team 1 · Guided by Prof. Dr.-Ing. Hamidreza Heidari · 20 Oct 2026, revised 4 Nov 2026 · github.com/bodhi584/aoi-false-call-classification

INTRODUCTION

Too many false alarms after automated inspection

99 %

of the solder joints flagged by AOI are false calls

Each one is checked again by hand at the manual inspection station.

Source: Siemens SMT line · 440,274 flagged joints, 39,742 boards, 132 days · Pfab et al. (2024), Data in Brief 53, 110110

Core paper: Pfab & Rothering (2025)

A second check trained once meets the targets on a test set, but misses 5–15 % of the true defects on later production periods.

A current problem: Siemens and NVIDIA, Erlangen, 21 Oct 2026

“AI second check of AOI results”: real defect or false call? Target: about 40 % fewer false calls.

Our goal

On later production periods: at most 1 % of true defects missed, at least 40 % of false calls released.

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ONE GOAL: AT MOST 1 % OF TRUE DEFECTS MISSED, AT LEAST 40 % OF FALSE CALLS RELEASED

Input → System → Output

INPUT

AOI data in time order

flagged solder joints: 70 AOI measurements + 5 categories; label from manual inspection

SYSTEM (OURS)

Classifier + 2 strategies

random forest or balanced random forest

decision threshold selection + concept drift adaptation (retraining)

OUTPUT PER LATER PERIOD

False calls released

≥ 40 % (manual checks saved)

Missed true defects

≤ 1 % (hard quality limit)

Key factors

Threshold

a threshold set once cannot hold 1 %

Drift

defect share changes: 0.09–3.11 % per period

Labels

released joints get no manual label

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PROJECT SCOPE

What we will and will not do

Aim

This project tests whether decision threshold selection and retraining make a second check of AOI results usable in production: at most 1 % of true defects missed and at least 40 % of false calls released on later periods.

Includes

- Baseline: reimplementation of Pfab & Rothering (2025)
- Threshold methods and retraining strategies
- Design of experiments: 72 settings × 10 seeds
- Simulation tool with threshold and retraining as settings

Out of scope (current phase)

- Real hardware deployment
- Integration with physical AOI machines
- New models or feature engineering (held as in the core paper)

Conditions and limitations

- One Siemens line, 132 days
- Measurements anonymised (no physical names)
- Labels from operators may contain errors
- Released joints get no label in production

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What the project will achieve, and how we measure it

1	Reimplement the method of Pfab & Rothering (2025) as the baseline and reproduce its problem on later periods.
2	Decision threshold selection: Neyman–Pearson threshold and prior shift adjustment.
3	Concept drift adaptation: periodic and drift-triggered retraining, on all data or a sliding window.
4	Design of experiments: 72 settings × 10 seeds against the baseline; find the setting that releases the most false calls within 1 %.

Requirements on every later period

≤ 1 % — missed true defects (FNR): hard quality limit

≥ 40 % — false calls released (TNR) = manual checks saved; expected 40 % to 80–90 %

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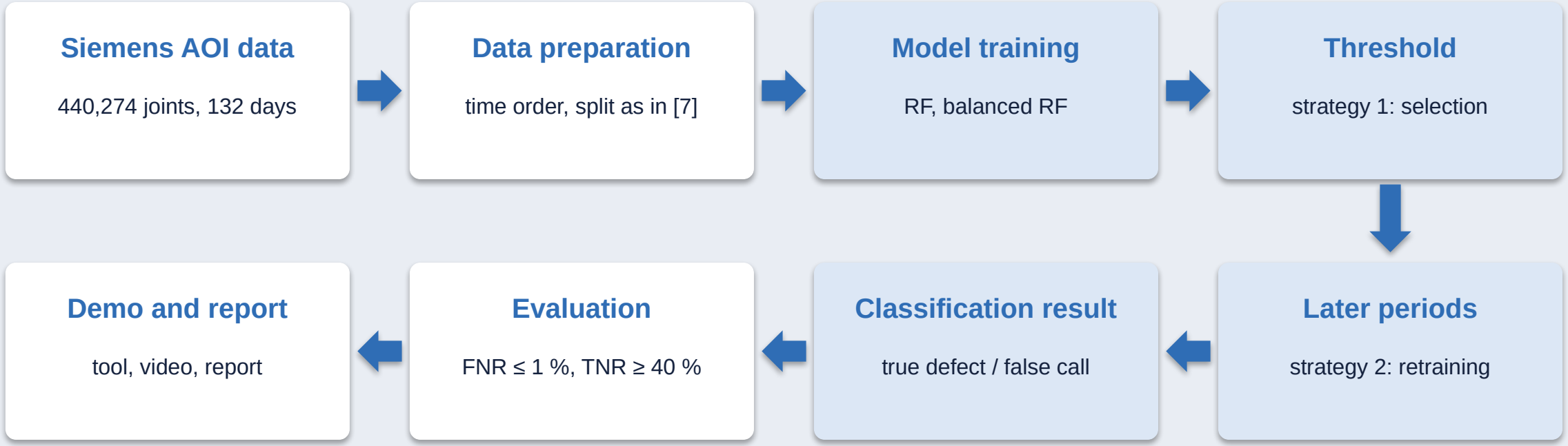
TECHNOLOGY STACK

Data science tools from the lecture (p. 22)

Process	Tool / Framework	Purpose
Data preparation	Python, pandas	Load the Siemens AOI data, keep time order, split as in Pfab & Rothering (2025)
Model training	scikit-learn, imbalanced-learn	Random forest, balanced random forest
Simulation	Python	Later periods in time order with threshold and retraining strategies
Evaluation	scikit-learn, pandas	Missed true defects (FNR), false calls released (TNR)
Visualization	matplotlib, Jupyter	Result figures, data notebook
Code sharing and tests	GitHub, pytest, GitHub Actions	Repository, tests and CI on every push

PROJECT FLOW CHART

From the Siemens data to the KPIs



Blue boxes: the system we build. White boxes: data and project steps.

WORKING METHODOLOGY

CRISP-DM in four steps

Step 1 · Data preparation

Siemens AOI data: 440,274 flagged joints, label from manual inspection

70 measurements + 5 one-hot categories; timestamp only for order

First half: 80 % training, 20 % test; second half: 5 later periods

Step 2 · Model development

Reimplemented the random forest and balanced random forest of Pfab & Rothering

Threshold at 99 % recall, taken from the best model

Baseline reproduces the problem of the paper

Step 3 · Adaptation strategies

Decision threshold selection: Neyman–Pearson, prior shift

Concept drift adaptation: periodic, drift-triggered retraining

Each period classified with data from earlier periods only

Step 4 · Evaluation and comparison

Missed true defects and false calls released per later period

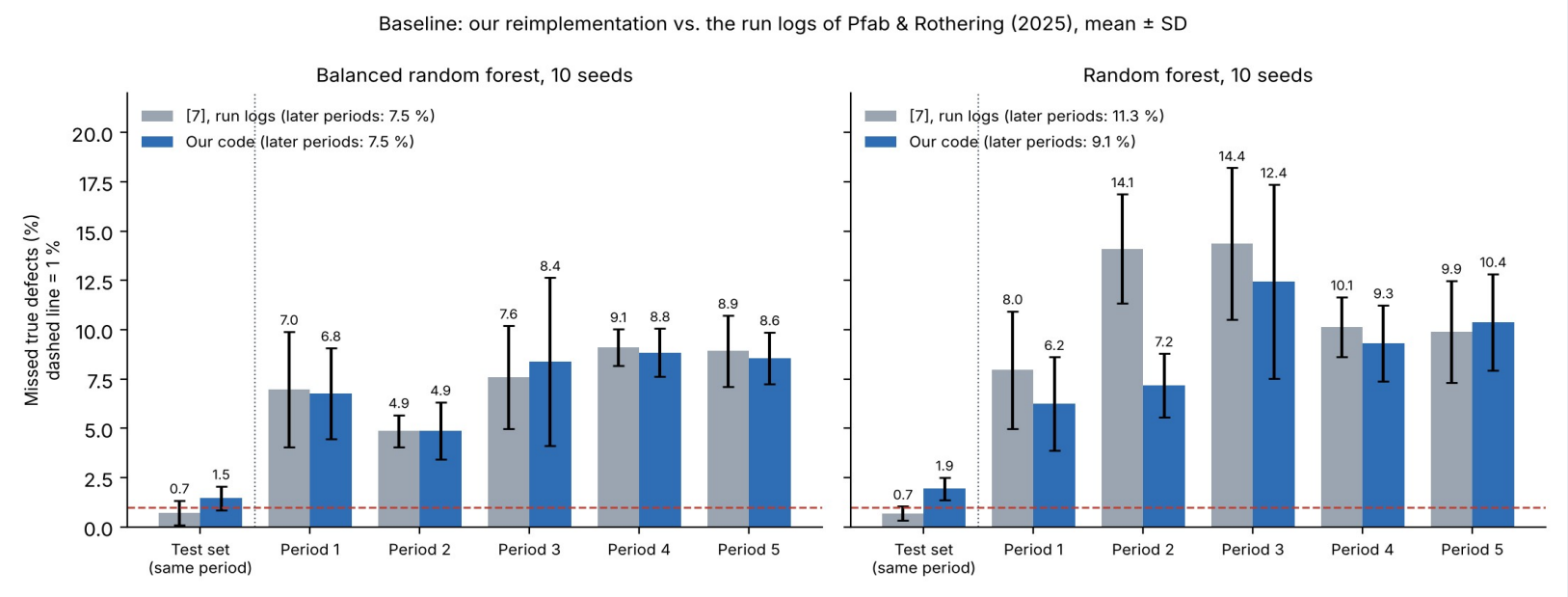
Design of experiments: 72 settings × 10 seeds vs. baseline

Simulation tool, demo video and report

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MVP · BASELINE

Our code shows the same problem as the core paper



What it shows

Test set: 1.5–1.9 % of true defects missed (paper: 0.7 %).

Later periods: 7.5–9.1 % missed (paper: 7.5–11.3 %), far above 1 %.

Next (P3): threshold methods and retraining against this baseline.

BILL OF MATERIALS (BOM)

What the project needs

Component	Specification	Quantity	Cost
Data set	Siemens AOI data, Mendeley (CC BY-NC 3.0), 333 MB	1	Free
Computer	Normal laptop or Google Colab (no graphics card needed)	4	Free
Software	Python and open-source libraries (scikit-learn, imbalanced-learn)	–	Free
Code hosting	GitHub	1 repository	Free

Total cost: 0 €

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EXPECTED OUTCOMES

What we hand in

- ✓ Core code: reimplementation of Pfab & Rothering (2025) with threshold and retraining strategies
- ✓ Simulation of later production periods: missed true defects and released false calls
- ✓ Results of the design of experiments: which threshold method and retraining strategy work
- ✓ Simulation tool with threshold and retraining as settings
- ✓ Demo video, run instructions, report, public GitHub repository

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SAFETY & ETHICAL CHECKLIST

Keeping it safe and realistic

- ✓ Simulation-only environment; no physical machinery risk
- ✓ Dataset is anonymized by Siemens (no personal data); used under its licence
- ✓ Ethical principles followed: transparency, fairness, accountability
- ✓ Avoid dataset bias (time-ordered split) and ensure reproducibility of results
- ✓ Safety first in the design: at most 1 % of true defects missed, as a hard limit

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FUTURE WORK

From the public data set to a real line

1 · Industry partner

An electronics plant shares AOI and manual-inspection records.

2 · AOI interface

Read each board's AOI result from the machine; send the classification to the manual inspection station.

3 · Shadow mode

The classifier runs next to the operators and only logs its results.

4 · Approval

Quality engineers compare the logs before any board skips manual inspection.

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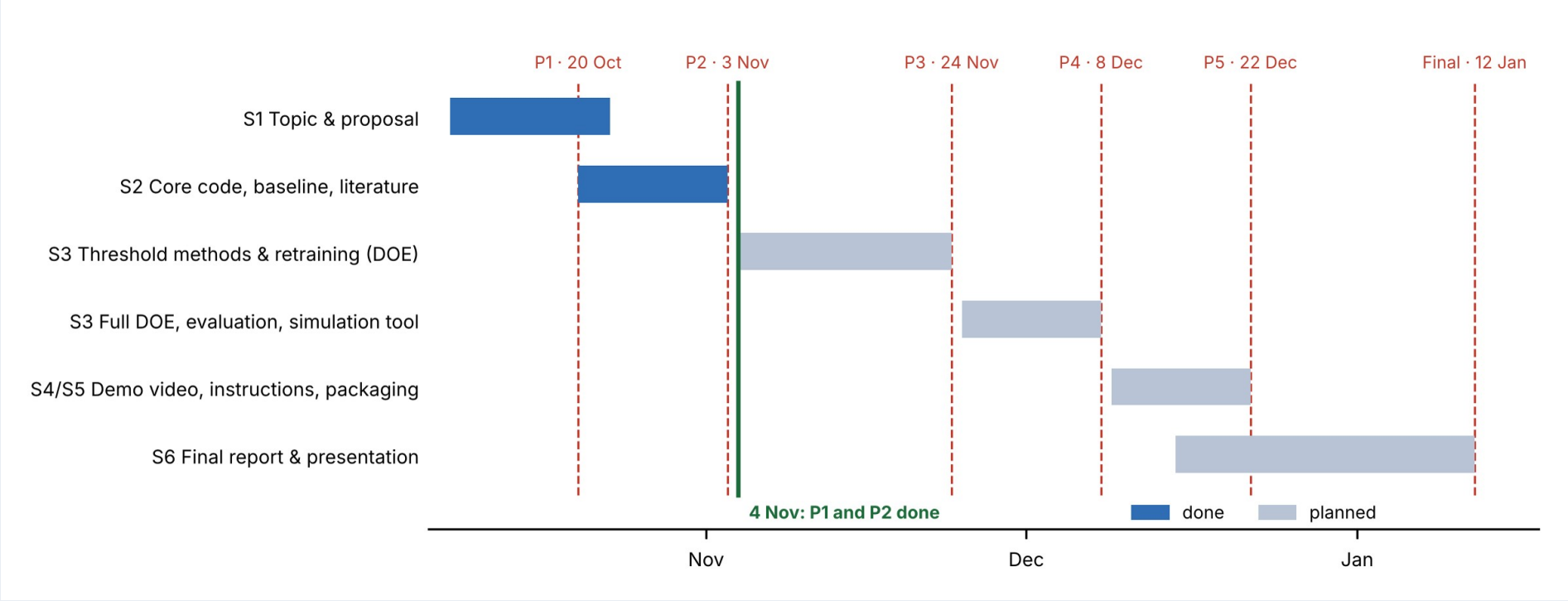
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MILESTONES

Software & Simulation project, S1–S6 (lecture p. 8, 33)



Repository link

github.com/bodhi584/aoi-false-call-classification

Thank you

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